

DEFINITION — HOW DATA IS DESCRIBED

CONTRACTS6



ODCS
BITOL @ LF



OpenAPI
LF



AsyncAPI
LF



GraphQL
LF



gRPC
CNCF



OData
OASIS

DATA PRODUCTS4



ODPS
BITOL @ LF



DPDS
ODM



ODPSpec
LF



DPROD
OMG

SCHEMA7



XML Schema
W3C



JSON Schema
IETF



SQL DDL
ISO/IEC



AVRO Schema
ASF



Protobuf
GOOGLE



LinkML
LINKML



Table Schema
FRICITIONLES S

SEMANTICS8



OSI
OSI INITIATIVE



RDF/OWL
W3C



DCAT
W3C



SKOS
W3C



SHACL
W3C



JSON-LD
W3C



schema.org
W3C



ShEx
W3C

STORAGE — WHERE DATA LIVES

FILE FORMATS8



CSV
IETF



JSON
IETF



XML
W3C



PARQUET
ASF



AVRO
ASF



ORC
ASF



Lance
LANCE



YAML
YAML.ORG

OPEN TABLE FORMATS4



Iceberg
ASF



Delta
LF



Hudi
ASF



Lance
LANCE

STORAGE SYSTEMS2



S3
AWS



HDFS
ASF

MOVEMENT — HOW DATA FLOWS BETWEEN SYSTEMS

DATABASE CONNECTIVITY4



JDBC
JCP



ODBC
ISO/IEC



ADBC
ASF



XMLA
MICROSOFT

INTERCONNECTION3



HTTP
IETF



FTP / SFTP
IETF



Delta Sharing
LF

MESSAGING5



Kafka
ASF



MQTT
OASIS



CloudEvents
CNCF



AMQP
OASIS



JMS
JAKARTA EE

TRANSFORMATION — HOW DATA IS PROCESSED AND RESHAPED

IN-MEMORY FORMAT2



DataFrame API
DATA-APIS



Apache Arrow
ASF

PROCESSING7



dbt
DBT LABS



Spark
ASF



Beam
ASF



Pandas
NUMFOCUS



Ibis
IBIS



SQL DML
ISO/IEC



XSLT
W3C

DISCOVERY — HOW DATA IS FOUND AND TRACED

CATALOG APIS5



Iceberg Catalog
ASF



Unity Catalog
LF



Hive Metastore
ASF



Schema Registry
CONFLUENT



DuckLake
DUCKDB LABS

LINEAGE2



OpenLineage
LF



PROV
W3C

OPERATIONS — HOW DATA IS QUERIED, OBSERVED, GOVERNED

QUERY5



SQL
ISO/IEC



SPARQL
W3C



Substrait
LF



GQL
ISO/IEC



MDX
MICROSOFT

DATA QUALITY3



dbt
DBT LABS



Great Expectations
GX LABS



SodaCL
SODA

OBSERVABILITY2



Open Telemetry
CNCF



OORS
BITOL @ LF

POLICIES2



OPA
CNCF



ODRL
W3C

AI INTERFACES2



MCP
LF



A2A
LF